

MUHAMMAD ZAKUAN BIN MOHD NOOR

Penang, Malaysia | contact@muzamono.com | github.com/muzamono

SUMMARY

Data engineer transitioning from a technical background in quality engineering and technical support, with self-built, deployed pipeline and full-stack projects. Experienced designing and running production-style ETL pipelines on Apache Airflow, PostgreSQL, Docker, and AWS, and building full-stack applications with FastAPI and React. Brings 5+ years of structured troubleshooting, root cause analysis, and cross-functional technical communication from engineering and support roles, now applied to building and operating data infrastructure.

DATA ENGINEERING PROJECTS

Stocks Tracker ETL | github.com/muzamono/stocks-tracker

- Built an end-to-end stock price ETL pipeline in Python, orchestrated with Apache Airflow, pulling daily data from the Alpha Vantage API into PostgreSQL.
- Deployed and scheduled the pipeline on AWS (EC2, RDS) using Docker Compose, with a daily cron-driven production run and an Elastic IP to eliminate recurring connectivity failures caused by IP changes on instance restart.
- Connected Metabase for downstream visualization and validated data integrity across local and cloud environments.

Receipt Tracker | github.com/muzamono/receipt-tracker

- Designed and built a full-stack expense tracking application: FastAPI backend, PostgreSQL database (Dockerized), and a React/Vite frontend with a mobile-first interface.
- Implemented receipt photo upload with HEIC conversion, CSV export, and monthly spend breakdowns with category visualization.
- Currently building an AI-assisted receipt parsing pipeline (text extraction using vision-model LLM).

PROFESSIONAL EXPERIENCE

Customer Care Advisor, Academic & Government Division | *Clarivate (Malaysia) Sdn. Bhd.* | Dec 2024 – Present

- Provide technical support to APAC enterprise customers via Salesforce, investigating platform issues by replicating problems and analyzing system behavior.
- Own support tickets end to end, coordinating with engineering and product teams to resolve complex, cross-functional issues within SLA.
- Analyze case trends and contribute to internal reporting using Excel and CRM data; awarded Clarivate Customer Champion (June 2025) for a 100% Customer Satisfaction Score.

Engineer, Quality Control (High Voltage Transmission Products) | *Mitsubishi Electric Co.* | Jan 2024 – Jun 2024

- Supported product testing and validation against technical specifications, applying structured troubleshooting and root cause analysis to identify product issues.
- Facilitated bilingual (Japanese/English) technical communication between international clients and internal teams.

Engineer, Quality Assurance (Steel Wire Products) | *Neturen Co.* | Apr 2021 – Jan 2024

- Conducted root cause analysis on production and quality issues, implementing corrective actions and driving process optimization.
- Collaborated with suppliers and internal teams to resolve technical problems, applying data-driven analysis to quality decisions.

Researcher (Master's Program) | *Neturen Co. / Sophia University* | Apr 2019 – Mar 2021

- Conducted research on material failure mechanisms involving data analysis and experimental validation; presented findings at COM 2020 (Canada).

TECHNICAL SKILLS

Languages & Data: Python, SQL

Data Engineering: Apache Airflow, PostgreSQL, ETL pipeline design, Docker, Docker Compose

Cloud & Infra: AWS (EC2, RDS), Linux/Unix, networking fundamentals (cURL, traceroute, netcat)

Application Development: FastAPI, React, REST APIs

Visualization & Tools: Metabase, Power BI, Excel, Git/GitHub, Salesforce CRM

Languages: English (TOEIC 990), Japanese (JLPT N1), Malay (Native)

EDUCATION

Sophia University, Tokyo, Japan — M.S. Mechanical Engineering (2019–2021), B.S. Mechanical Engineering (2017–2019)

UniKL IPROM, Malaysia — Diploma in Mechanical Engineering (2014–2017), CGPA 3.99, Dean's List, Overall Best Student